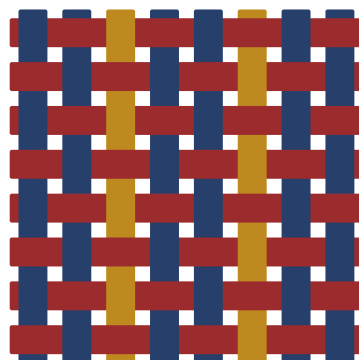




Nonlinear Algebra

Fill-in notes · Chapters 1--2 · Michałek & Sturmfels

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dyed in indigo, madder & weld; ruled in iron-gall. woven on LOOM.

THE DICTIONARY, AND HOW TO READ THESE NOTES

□□□□

The whole of Chapters 1--2 is one idea, stated twice by Sophie Germain: ``**Algebra is but written geometry**'' (Ch. 1) and ``**Geometry is but drawn algebra**'' (Ch. 2). There is a dictionary between the *algebra* side (ideals in the polynomial ring $K[\mathbf{x}]$) and the *geometry* side (varieties in K^n or $\mathbb{P}^n$). Learn the dictionary and the two chapters become one.

How to read these notes (passive and active)

- **Read** the strands and the stated theorems. That is the passive intake.
- **Fill** the blanks _____, the [fill in: the missing step] gaps inside proofs, and the *Your turn* boxes. That is the active recall. The book is your answer key.
- Bump each section's opening \warmth{0} to 0--5 once you can rebuild it cold.

The master dictionary (fill it in as you go)

Algebra: ideals in $K[\mathbf{x}]$	Geometry: varieties in $K^n, \mathbb{P}^n$
ideal I	variety $\mathcal{V}(I)$ (§4)
vanishing ideal $\mathcal{I}(W)$ (radical)	variety $W = \mathcal{V}(\mathcal{I}(W))$
$I \subseteq J$	$\mathcal{V}(I)$ _____ $\mathcal{V}(J)$ (which way?)
sum $I + J$	_____ of $\mathcal{V}(I), \mathcal{V}(J)$
intersection $I \cap J$	_____ of $\mathcal{V}(I), \mathcal{V}(J)$
prime ideal	_____ variety (§4)
maximal ideal $\langle x_1 - p_1, \dots, x_n - p_n \rangle$	a single point _____
radical ideal	($\mathcal{I} = \mathcal{V}$ inverse, up to radical: Nullstellensatz, Ch. 6)
quotient $K[\mathbf{x}]/\mathcal{I}(W)$	coordinate ring $K[W] =$ functions on W
dimension of I (§3)	dimension of $\mathcal{V}(I)$
degree of I	# points of a general linear slice
$\langle f, \partial f / \partial x_1, \dots \rangle$ has no zeros	$\mathcal{V}(f)$ is _____
homogeneous ideal	projective variety (§5)
primary decomposition (Ch. 3)	decomposition into irreducibles

The map $\mathcal{V}$ sends algebra to geometry; the map $\mathcal{I}$ sends geometry back to algebra. They are *inclusion-reversing*, and they are almost inverse to each other. ``Almost'' is the content of the Nullstellensatz (Chapter 6); for now, $\mathcal{I}(\mathcal{V}(I)) = \text{rad}(I)$ is the slogan to carry.

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1 IDEALS

□□□□ The algebra side. An ideal is the algebraic shadow of a geometric shape.

The objects of nonlinear algebra live in the **polynomial ring** $K[\mathbf{x}] = K[x_1, \dots, x_n]$ over a field K (favorite: $K = \mathbb{Q}$; also $\mathbb{R}, \mathbb{C}, \overline{\mathbb{K}}, \mathbb{F}_q$). A polynomial is a finite K -combination of monomials $\mathbf{x}^{\mathbf{a}} = x_1^{a_1} \cdots x_n^{a_n}$, and its **degree** is $\max\{|\mathbf{a}| : c_{\mathbf{a}} \neq 0\}$. The hero of the book is one cubic surface; meet it now, it returns everywhere.

Example 1.1 (the running cubic, $n = 3$).

$$f = \det \begin{pmatrix} 1 & x & y \\ x & 1 & z \\ y & z & 1 \end{pmatrix} = 2xyz - x^2 - y^2 - z^2 + 1.$$

$\mathcal{V}(f) \subset \mathbb{R}^3$ is a cubic surface (Cayley's nodal cubic). Its four **singular points** are the common zeros of f and its three partials $\partial f / \partial x = 2yz - 2x$, etc.

Your turn. Solve $2yz - 2x = 2xz - 2y = 2xy - 2z = 0$ together with $f = 0$. Check the four singular points are $(1, 1, 1), (1, -1, -1), (-1, 1, -1), (-1, -1, 1)$.

? Why must a singular point also satisfy $f = 0$ here? (Euler's identity for homogeneous-
ish f -- or just check.)

What an ideal is

Definition 1.2 (ideal, Def. 1.1). All rings here are commutative with 1. A nonempty subset $I \subseteq R$ is an **ideal** if

- (a) $f \in R$ and $g \in I \Rightarrow \underline{\hspace{2cm}} \in I$ (absorbs multiplication);
- (b) $f, g \in I \Rightarrow \underline{\hspace{2cm}} \in I$ (closed under addition).

Equivalently, $I = \ker \phi$ for some ring homomorphism $\phi : R \rightarrow S$.

Given $\mathcal{F} \subseteq R$, the smallest ideal containing it is $\langle \mathcal{F} \rangle$, the ideal **generated** by $\mathcal{F}$ (all polynomial-coefficient combinations). The quotient R/I is a ring (Prop. 1.2), built from congruence classes $f + I$.

? Why is $\{\text{even integers}\}$ an ideal of $\mathbb{Z}$? Which homomorphism has it as kernel?

Proposition 1.3 (four operations, Prop. 1.3). If $I, J \subseteq R$ are ideals, so are the sum $I + J$, the intersection $I \cap J$, the product $IJ = \langle fg : f \in I, g \in J \rangle$, and the colon (quotient) $(I : J) = \{f \in R : fJ \subseteq I\}$.

Proof. We check $(I : J)$ is an ideal. For (a): $f \in R, g \in (I : J) \Rightarrow (fg)J = f(gJ) \subseteq fI \subseteq I$. For (b): $f, g \in (I : J) \Rightarrow (f + g)J \subseteq fJ + gJ \subseteq I + I = I$. [fill in: do the same two checks for $I \cap J$]

Your turn. ($n = 1$, Example 1.4.) In $\mathbb{Q}[x]$ let $I = \langle (x+2)^3 \rangle$ and $J = \langle (x-1)(x+2) \rangle$. Compute, as principal ideals (factor first!):

$$I \cap J = \underline{\hspace{2cm}}, \quad IJ = \underline{\hspace{2cm}},$$

$$I + J = \underline{\hspace{2cm}}, \quad (I : J) = \underline{\hspace{2cm}}.$$

Moral: *arithmetic in $\mathbb{Q}[x]$ is just like arithmetic in $\mathbb{Z}$ ($\cap = \text{lcm}$, $+$ = gcd).*

Four kinds of ideal, read off the quotient

Definition 1.4 (the property dictionary). For an ideal $I \subseteq R$ and its quotient R/I :

I is...	definition	...iff R/I is
maximal	no proper ideal contains I	a <u> </u>
prime	$fg \in I \Rightarrow f \in I$ or $g \in I$	an <u> </u>
radical	$f^s \in I \Rightarrow f \in I$	free of <u> </u> elements
primary	$fg \in I, g \notin I \Rightarrow f^s \in I$	one where zero divisors are nilpotent

Proposition 1.5 (implications, Prop. 1.6). I maximal $\Rightarrow$ prime $\Rightarrow$ radical, and prime $\Rightarrow$ primary. None reverses; radical + primary = prime; any intersection of primes is radical.

Proof. Maximal $\Rightarrow$ prime since a field has no zero divisors. Prime $\Rightarrow$ radical: take $g = f^{s-1}$ and induct on s . [fill in: prove radical + primary $\Rightarrow$ prime (assume $fg \in I, f \notin I$; use primary then radical)]

Back to the cubic (Example 1.1). Its singular locus is $\mathcal{J} = \langle yz-x, xz-y, xy-z \rangle$, a *radical* ideal that factors as an intersection of **five** maximal ideals; by the Chinese Remainder Theorem $\mathbb{R}[x, y, z]/\mathcal{J} \cong \mathbb{R}^5$. The number five is the five complex points of $\mathcal{V}(\mathcal{J})$, four of them the real nodes. Algebra (an intersection of maximal ideals) has just *counted geometry* (a finite set of points). That is the dictionary firing for the first time.

? Match the non-reversing counterexamples in $\mathbb{R}[x, y]$: $\langle x^2 \rangle$, $\langle x(x-1) \rangle$, $\langle x \rangle$ to "primary not radical", "radical not primary", "prime not radical". **Exercise 1.8** (book): identify the maximal, prime, radical, primary ideals of $R = \mathbb{Z}$. Same four words, older ring.

2 GRÖBNER BASES

□□□□ The engine. Gröbner is to ideals what Gaussian elimination is to matrices.

An ideal has many generating sets and no canonical basis. **Gröbner bases** fix this: they are the right generating set once you pick a *monomial order*. Think of them as the Euclidean algorithm for $n \geq 2$ variables, or Gaussian elimination for degree ≥ 2 . The book's creed: **nonlinear algebra is the next step after linear algebra.**

Monomial orders and initial terms

Definition 2.1 (monomial order, Def. 1.10). A total order $<$ on $\mathbb{N}^n$ (monomials) is a **monomial order** if $(0, \dots, 0) \preceq \mathbf{a}$ for all $\mathbf{a}$, and $\mathbf{a} \preceq \mathbf{b} \Rightarrow \mathbf{a} + \mathbf{c} \preceq \mathbf{b} + \mathbf{c}$. Three standards (all with $x_1 \succ \dots \succ x_n$): *lex*, *deglex* (degree first, then lex), *degrevlex* (degree first, then *smallest* last variable wins). Any monomial order refines divisibility: $\mathbf{x}^{\mathbf{a}} \mid \mathbf{x}^{\mathbf{b}} \Rightarrow \mathbf{a} \preceq \mathbf{b}$.

The **initial monomial** $\text{in}_{<}(f)$ is the $<$ -largest monomial of f .

Your turn. Let $f = x^2 + xz^2 + y^3$ with $x \succ y \succ z$. Fill the three initial monomials:

$$\text{in}_{\text{lex}}(f) = \underline{\hspace{2cm}}, \quad \text{in}_{\text{deglex}}(f) = \underline{\hspace{2cm}}, \quad \text{in}_{\text{degrevlex}}(f) = \underline{\hspace{2cm}}.$$

? For degrevlex, why does y^3 beat xz^2 ? Both have degree 3 - compare the last variable.

Finiteness: Dickson and the Basis Theorem

Theorem 2.2 (Dickson's Lemma, Thm. 1.8). Any infinite subset of $\mathbb{N}^n$ contains a pair $\{\mathbf{a}, \mathbf{b}\}$ with $\mathbf{a} \leq \mathbf{b}$ (coordinatewise). Equivalently: any nonempty $\mathcal{M} \subseteq \mathbb{N}^n$ has a *finite* set of coordinatewise-minimal elements.

Proof. Induction on n . For $n = 1$, any two naturals are comparable. [fill in: induction step: slice $\mathcal{M}$ by the last coordinate i , set $\mathcal{M}_i \subseteq \mathbb{N}^{n-1}$; if some $\mathcal{M}_i$ is infinite use IH, else ...]

For an ideal I , the **initial ideal** is $\text{in}_{<}(I) = \langle \text{in}_{<}(f) : f \in I \setminus \{0\} \rangle$, a monomial ideal. A finite $\mathcal{G} \subseteq I$ with $\text{in}_{<}(I) = \langle \text{in}_{<}(g) : g \in \mathcal{G} \rangle$ is a **Gröbner basis** (it exists, by Dickson).

Theorem 2.3 (a Gröbner basis generates, Thm. 1.13). If $\mathcal{G}$ is a Gröbner basis of I , then $I = \langle \mathcal{G} \rangle$.

Proof. Suppose not. Pick $f \in I \setminus \langle \mathcal{G} \rangle$ with $\text{in}_{<}(f) = \mathbf{x}^{\mathbf{b}}$ minimal. Some $g \in \mathcal{G}$ has $\text{in}_{<}(g) \mid \mathbf{x}^{\mathbf{b}}$, say $\mathbf{x}^{\mathbf{b}} = \mathbf{x}^{\mathbf{c}} \text{in}_{<}(g)$. [fill in: form $f - \mathbf{x}^{\mathbf{c}}g$: it lies in I , has strictly smaller initial monomial, contradiction]

Corollary 2.4 (Hilbert Basis Theorem, Cor. 1.14). Every ideal in $K[\mathbf{x}]$ is $\underline{\hspace{2cm}}$.

Reduced Gröbner bases and standard monomials

Definition 2.5 (reduced, Def. 1.15). $\mathcal{G}$ is a **reduced** Gröbner basis if (a) each leading coefficient is 1, and (b) for distinct $g, h \in \mathcal{G}$, no monomial of g is a multiple of $\text{in}_{<}(h)$.

Theorem 2.6 (uniqueness + a basis, Thm. 1.16–1.17). Fix I and $<$. Then I has a *unique* reduced Gröbner basis. Moreover the **standard monomials** $\mathcal{S}_{<}(I) = \{\text{monomials} \notin \text{in}_{<}(I)\}$ form a K -vector-space basis of the quotient $K[\mathbf{x}]/I$.

Proof of the basis claim. Independence: a nonzero f with image 0 in $K[\mathbf{x}]/I$ lies in I , so $\text{in}_<(f) \in \text{in}_<(I)$ is *not* standard. Spanning: [fill in: take a minimal $\mathbf{x}^c$ not in the span; it lies in $\text{in}_<(I)$, divide, contradiction]

Your turn. (Symmetric polynomials, from Example 1.18.) The elementary symmetric polynomials $\mathcal{F} = \{x + y + z, xy + xz + yz, xyz\}$ in $\mathbb{Q}[x, y, z]$ have reduced Gröbner basis (lex) $\mathcal{G} = \{\underline{x} + y + z, \underline{y}^2 + yz + z^2, \underline{z}^3\}$.

- List the _____ standard monomials $\mathcal{S}_<(I)$ (they index a basis of $\mathbb{Q}[x, y, z]/I$).
- This quotient is the regular representation of which group? _____

? # standard monomials
 $= \dim_K K[\mathbf{x}]/I =$
 # solutions counted with multiplicity.
 Here $= |\mathcal{S}_3| = 6$.

Buchberger's algorithm computes the reduced Gröbner basis from any $\mathcal{F}$. Two warnings the book stresses: the **choice of order matters enormously** (Example 1.19: two random quartics in $\mathbb{P}^3$ give 5 generators of degree ≤ 7 in degrevlex, but 150 of degree ≤ 73 in lex), and **lex is the order for elimination** (Chapter 4). Run a computer algebra system as you read.

■ 3 DIMENSION AND DEGREE

□□□□ Two numbers attached to an ideal, both squeezed out of one counting series.

The two most important invariants of an ideal are its **dimension** and **degree**. Both are read off a single generating function that *counts monomials not in the ideal*, the Hilbert series. The definition here is purely combinatorial; §4 reveals that it matches the geometric dimension and degree of the variety.

Exercise 1
 (book): an ideal in $K[\mathbf{x}]$ is principal $\iff$ its reduced Gröbner basis is a singleton. (One direction is easy; the other uses uniqueness.)

Counting what is left

Definition 3.1 (*Hilbert function & series, Def. 1.20–1.21*). For a monomial ideal $I \subset K[\mathbf{x}]$, the **Hilbert function** $h_I(q)$ is the number of monomials of degree q that are _____ I . Its generating function is the **Hilbert series** $\text{HS}_I(z) = \sum_{q \geq 0} h_I(q) z^q$.

Your turn. Warm up by counting monomials directly.

- Zero ideal $I = \{0\}$: every monomial survives, so $h_{\{0\}}(q) = \binom{n+q-1}{n-1}$ and $\text{HS}_{\{0\}}(z) = \underline{\hspace{2cm}}$.
- Principal $I = \langle \mathbf{x}^a \rangle$ of degree e : subtract the multiples, giving $\text{HS}_I(z) = \frac{1 - z^e}{(1 - z)^n}$. Read off: $\dim = \underline{\hspace{2cm}}$, $\deg = \underline{\hspace{2cm}}$.

? This h is a polynomial in q of degree $n - 1$.

Theorem 3.2 (shape of the series, Thm. 1.25). For any monomial ideal $I \subset K[\mathbf{x}]$,

$$\text{HS}_I(z) = \frac{\kappa_I(z)}{(1-z)^n}, \quad \kappa_I \in \mathbb{Z}[z], \kappa_I(0) = 1,$$

and there is a **Hilbert polynomial** HP_I with $\text{HP}_I(q) = h_I(q)$ for all large q .

Proof. Inclusion-exclusion over the minimal generators $m_1, \dots, m_r$. For $\tau \subseteq \{1, \dots, r\}$ let $e_\tau = \deg \text{lcm}(m_i : i \in \tau)$. Then $\kappa_I(z) = \sum_{\tau} (-1)^{|\tau|} z^{e_\tau}$. [fill in: regroup to get $h_I(q) = \sum_{\tau} (-1)^{|\tau|} \binom{n+q-e_\tau-1}{n-1}$; check it is a polynomial for $q \gg 0$]

Reading off dimension and degree

Definition 3.3 (dimension & degree, Def. 1.28). Write $\text{HP}_I(q) = \frac{g}{(d-1)!} q^{d-1} +$ (lower order). Then the **dimension** of I is d and the **degree** is the positive integer g . If $\text{HP}_I \equiv 0$ then $\dim = 0$ and $\deg I = \dim_K K[\mathbf{x}]/I$ (a finite number).

Your turn. Two computations the book recommends.

- (Ex. 1.31) Principal ideal generated by a single monomial of degree e : confirm $\dim = n-1$ and $\deg = e$ from $\text{HP}_I(q) = \frac{e}{(n-2)!} q^{n-2} + \dots$.
- (Ex. 1.32) $n = 2m$, $I = \langle x_1 x_2, x_3 x_4, \dots, x_{2m-1} x_{2m} \rangle$: find $\dim I = \underline{\hspace{2cm}}$ and $\deg I = \underline{\hspace{2cm}}$.

? Slogan: dimension is the *growth rate* of h_I ; degree is its *leading coefficient* (times $(d-1)!$).

? Try $m = 2$:
 $I = \langle x_1 x_2, x_3 x_4 \rangle$.
The degree is a power of two.

Why these two numbers (geometry preview, Remark 1.36 / Example 1.37). For the variety $\mathcal{V}(I)$: $\dim = 0$ iff it is finitely many points, and then $\deg =$ the *number* of points (counted by the radical). A curve has $\dim 1$, a surface $\dim 2$. The degree measures how curvy the shape is. A clean fact: a prime ideal has degree 1 iff it is generated by linear polynomials. For the running cubic, $\langle f \rangle$ has $\dim 2$, $\deg 3$; its singular locus has $\dim 0$, $\deg 5$; together with f , $\dim 0$, $\deg 4$ (the four nodes).

4 AFFINE VARIETIES

□□□□ The geometry side, and the bridge. "Geometry is but drawn algebra."

A **variety** is the solution set of polynomial equations, $\mathcal{V}(I) = \{p \in K^n : f(p) = 0 \forall f \in I\}$. Different ideals can give the same variety, so geometry sees only I up to radical. The reverse map $\mathcal{J}(W) = \{f : f|_W = 0\}$ sends a shape back to its (radical) ideal. These two

Exercise 14 (book): let X be a generic 2×2 matrix of variables, $I_s = \langle \text{entries of } X^s \rangle$. Investigate $\dim, \deg$ of I_s for $s = 2, 3, 4, \dots$. A small experiment with a CAS pays off.

maps are the dictionary; this section makes them precise.

The two maps $\mathcal{V}$ and $\mathcal{I}$

Definition 4.1 (*variety and vanishing ideal*). For an ideal $I \subseteq K[\mathbf{x}]$ and a subset $W \subseteq K^n$,

$$\mathcal{V}(I) = \{p \in K^n : f(p) = 0 \forall f \in I\}, \quad \mathcal{I}(W) = \{f \in K[\mathbf{x}] : f(p) = 0 \forall p \in W\}.$$

$\mathcal{I}(W)$ is always a **radical** ideal. A subset is a *variety* iff $W = \mathcal{V}(\mathcal{I}(W))$.

? Both maps reverse inclusions. Fill:
 $I \subseteq J \Rightarrow \mathcal{V}(I) \supseteq \mathcal{V}(J)$,
 and $V \subseteq W \Rightarrow \mathcal{I}(W) \supseteq \mathcal{I}(V)$

Irreducible = prime: the keystone

Definition 4.2 (*irreducible*). A variety is **irreducible** if it is not a finite union of proper subvarieties: $\mathcal{V}(I) = \mathcal{V}(J) \cup \mathcal{V}(J') \Rightarrow \mathcal{V}(I) = \mathcal{V}(J)$ or $\mathcal{V}(J')$.

Proposition 4.3 (*Prop. 2.3*). A variety $W \subseteq K^n$ is irreducible $\iff$ its ideal $\mathcal{I}(W)$ is prime.

Proof. ($\Leftarrow$) Suppose $\mathcal{I}(W)$ prime and $W = \mathcal{V}(J) \cup \mathcal{V}(J')$ with $W \neq \mathcal{V}(J)$. Pick $f \in J$, $v \in W$ with $f(v) \neq 0$, so $f \notin \mathcal{I}(W)$. For $g \in J'$, the product fg vanishes on $\mathcal{V}(J) \cup \mathcal{V}(J') = W$, so $fg \in \mathcal{I}(W)$; primality gives $g \in \mathcal{I}(W)$. Hence $W \subseteq \mathcal{V}(J')$. [fill in: ($\Rightarrow$) W irreducible: if $fg \in \mathcal{I}(W)$ then $W = (W \cap \mathcal{V}(f)) \cup (W \cap \mathcal{V}(g))$; conclude f or $g \in \mathcal{I}(W)$]

This is the dictionary's keystone: **prime ideals** $\leftrightarrow$ **irreducible varieties**, exactly as prime numbers are to $\mathbb{Z}$. In applications (algebraic statistics), a model is the image of a polynomial map; its Zariski closure is irreducible, so its ideal is prime, and is computed as the kernel of the dual ring map. The closed sets of the **Zariski topology** are precisely the varieties; the coordinate ring $K[W] = K[\mathbf{x}]/\mathcal{I}(W)$ is the ring of polynomial functions on W , and a map of varieties $f : W_1 \rightarrow W_2$ pulls back to a ring map $f^* : K[W_2] \rightarrow K[W_1]$ (a *contravariant* functor).

Dimension and degree, geometrically

Example 4.4 (*linear subspace, Ex. 2.15*). For a linear subspace $V \subseteq K^n$: $\dim V$ is its linear-algebra dimension, $\mathcal{I}(V) = \langle x_1, \dots, x_s \rangle$ with $s = n - \dim V$, and $\deg V = \underline{\hspace{1cm}}$.

? Degree 1 $\iff$ prime ideal generated by linear forms.

A method to get $\dim \mathcal{V}(I)$: compute $\text{in}_<(I)$; find the smallest variable set S , $|S| = d$, hitting every generator; then $\dim = n - d$.

Theorem 4.5 (*Bézout, Thm. 2.16*). Let $f_1, \dots, f_k$ be *general* polynomials of degrees $d_1, \dots, d_k > 0$ in n variables, and $I = \langle f_1, \dots, f_k \rangle$. Then

$$\dim(I) = \underline{\hspace{1cm}}, \quad \deg(I) = \underline{\hspace{1cm}}.$$

(Such I is a **complete intersection**; $\dim \geq n - k$ always, by Krull.)

Your turn. Two general conics in the plane ($n = 2, k = 2, d_1 = d_2 = 2$): Bézout predicts $\dim = \underline{\hspace{1cm}}$ and $\deg = \underline{\hspace{1cm}}$ – i.e. they meet in $\underline{\hspace{1cm}}$ points (over $\overline{K}$). Sketch two ellipses meeting in 4 points to see it.

Smooth vs singular

Definition 4.6 (*singular point*). $p \in \mathcal{V}(f)$ is **singular** if all partials vanish: $\partial f / \partial x_i(p) = 0$ for all i . The singular locus is $\mathcal{V}(\langle f, \partial f / \partial x_1, \dots, \partial f / \partial x_n \rangle)$; if empty, $\mathcal{V}(f)$ is **smooth**. For an irreducible Y of dimension d cut by a prime I , the point p is singular iff $\text{rank Jac}(p) < n - d$ (codimension).

5 PROJECTIVE VARIETIES

□□□□ Add points at infinity and the geometry gets simpler: parallel lines finally meet.

Projective space $\mathbb{P}^n = \mathbb{P}(K^{n+1})$ is the set of *lines through the origin* in K^{n+1} , with coordinates $[a_0 : \dots : a_n]$ defined up to scaling. It is covered by $n + 1$ affine charts $\{a_i \neq 0\} \cong K^n$, and over $\mathbb{R}, \mathbb{C}$ it is **compact**. Geometers prefer $\mathbb{P}^n$: parallel lines in K^2 do not meet, but any two lines in $\mathbb{P}^2$ do. The price is that we may only use **homogeneous** polynomials, since only their vanishing is well defined on a line.

Homogeneous ideals, cones, closures

Definition 5.1 (*projective variety*). For homogeneous $f_1, \dots, f_k$, $\mathcal{V}(f_1, \dots, f_k) = \{[a] \in \mathbb{P}^n : f_i(a) = 0 \ \forall i\}$. An ideal is **homogeneous** if generated by homogeneous polynomials. The **affine cone** $\hat{X} \subseteq K^{n+1}$ is cut by the same ideal, and

$$\dim(X) = \dim(\hat{X}) - \underline{\hspace{1cm}}, \quad \deg(X) = \deg(\hat{X}).$$

The **projective closure** $\bar{Y} \subseteq \mathbb{P}^n$ of an affine $Y \subseteq K^n$ has ideal $\bar{I}$ obtained by **homogenizing**, and Proposition 2.18 computes it cleanly: take a degree-compatible reduced Gröbner basis $\mathcal{G}$ of I and homogenize each $g \in \mathcal{G}$.

Theorem 5.2 (*invariants survive the closure, Cor. 2.19*). $\dim(\bar{Y}) = \dim(Y)$ and $\deg(\bar{Y}) = \deg(Y)$.

Your turn. (Rational normal curve, Ex. 2.20.) Let $I = \langle x_i - x_1^i : i = 2, \dots, n \rangle$, so $Y = \mathcal{V}(I)$ is the curve $t \mapsto (t, t^2, \dots, t^n)$. Its projective closure $\bar{Y} \subset \mathbb{P}^n$ has ideal generated by the 2×2 minors of

$$\begin{pmatrix} x_0 & x_1 & x_2 & \cdots & x_{n-1} \\ x_1 & x_2 & x_3 & \cdots & x_n \end{pmatrix}.$$

How many such minors are there? $\underline{\hspace{1cm}}$. The curve $\bar{Y}$ has degree $\underline{\hspace{1cm}}$ and is the

2. The four nodes of the Bertini (book): running cubic an ideal are exactly its generated by singular locus -- general the dim 0, deg 4 polynomials is ideal from §3. prime and its smoothness = variety is the tangent space, has the right dim > 0. So "random" equations cut out nice irreducible smooth varieties; degeneracy is special.

? Why is the cone $\hat{X}$ singular at the origin once $\deg \geq 2$? Every line of the cone passes through 0.

rational normal curve.

? The $\binom{n}{2}$ minors degree n . The initial ideal is $\langle x_1, \dots, x_{n-1} \rangle^2$.

Why projective: intersections always behave

Theorem 5.3 (projective intersection, Thm. 2.22). Let K be algebraically closed and $X, Y \subseteq \mathbb{P}^n$ projective varieties with $d_1 = \dim X$, $d_2 = \dim Y$. Then

$$\dim(X \cap Y) \geq \underline{\hspace{2cm}}.$$

In particular, if $d_1 + d_2 \geq n$ then $X \cap Y$ is $\underline{\hspace{2cm}}$.

Your turn. (Ex. 2.23, why the hypotheses matter.) Two quadric surfaces $X, Y \subseteq \mathbb{P}^3$ ($n = 3$, $d_1 = d_2 = 2$).

- Theorem prediction: $\dim(X \cap Y) \geq 2 + 2 - 3 = \underline{\hspace{1cm}}$, and the intersection is nonempty.
- Over $\mathbb{C}$: their intersection is a union of four lines, $\dim = 1$. ☒
- Over $\mathbb{R}$: the book's pair meets in only *two points*, $\dim = 0$. Which hypothesis of the theorem just failed? $\underline{\hspace{2cm}}$

Many models in science are projective varieties given by homogeneous constraints. Two showcases: the **nilpotent $n \times n$ matrices** form an irreducible projective variety in $\mathbb{P}^{n^2-1}$ of $\dim n^2 - n - 1$ and $\deg n!$, a complete intersection cut by the non-leading coefficients of the characteristic polynomial (for $n = 2$: $\mathcal{J}(X) = \langle \text{trace}(A), \det(A) \rangle$); and the **Kalman varieties** of matrices with an eigenvector in a fixed subspace, basic in control theory. The lesson of the chapter: when constraints are homogeneous, regard the model as living in $\mathbb{P}^n$.

6 GEOMETRY IN LOW DIMENSIONS

□□□□ Where equations become manifolds: circles, Riemann spheres, ovals, and a torus.

Smooth projective varieties in low dimensions *are* familiar manifolds. Over $\mathbb{R}$ a smooth projective variety of dimension d is a compact real manifold of dimension d ; over $\mathbb{C}$, of dimension $2d$. So $\mathbb{P}_{\mathbb{R}}^1$ is a $\underline{\hspace{2cm}}$ and $\mathbb{P}_{\mathbb{C}}^1$ is the **Riemann sphere**. This section reads the topology straight off the algebra (degree, genus).

Curves in the plane: ovals, pseudolines, genus

Your turn. ($n = 1$, binary forms.) A proper subvariety of $\mathbb{P}_{\mathbb{C}}^1$ is the zero set of a binary form $f(x, y)$ (homogeneous in two variables). For $f = x^{11}y - 11x^6y^6 - xy^{11}$, the variety $\mathcal{V}(f)$ has $\dim 0$ and degree $\underline{\hspace{1cm}}$. These 12 points on the Riemann sphere are Klein's **icosahedron** vertices: 4 real, 8 in conjugate pairs.

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Exercise 17 (book):  $X \subset \mathbb{C}^4$  cut by two random cubics. What does Bézout tell you about  $\dim$ ,  $\deg$ , and what does Bertini tell you about smoothness and primeness?

? Degree of a binary form = its number of roots in  $\mathbb{P}_{\mathbb{C}}^1$ , with multiplicity.

For a curve  $C \subset \mathbb{P}_{\mathbb{R}}^2$ , a connected component is a **pseudoline** if removing it leaves  $\mathbb{P}_{\mathbb{R}}^2$  connected, and an **oval** otherwise (it has an inside and an outside, like a conic).

**Theorem 6.1** (*curves in the plane, Thm. 2.26*). Let  $C$  be a smooth curve of degree  $d$  in  $\mathbb{P}_{\mathbb{K}}^2$ .

- If  $K = \mathbb{C}$ :  $C$  is an orientable surface of **genus**  $g = \frac{(d-1)(d-2)}{2}$ .
- If  $K = \mathbb{R}$ :  $C$  has at most  $g + 1$  components; if  $d$  is even all are ovals, if  $d$  is odd exactly one is a \_\_\_\_\_.

**Your turn.** Fill the genus by degree:  $d = 1 \Rightarrow g = \underline{\hspace{1cm}}$ ,  $d = 2 \Rightarrow g = \underline{\hspace{1cm}}$ ,  $d = 3 \Rightarrow g = \underline{\hspace{1cm}}$ . So lines and conics are spheres ( $g = 0$ ), and a smooth cubic is a \_\_\_\_\_ ( $g = 1$ ).

### Elliptic curves: a cubic that is a torus

A smooth cubic ( $d = 3, g = 1$ ), e.g. the Weierstrass curve  $zy^2 = x^3 - xz^2$ , is an **elliptic curve**. Over  $\mathbb{R}$  it shows an oval and a pseudoline; over  $\mathbb{C}$  it is homeomorphic to a **torus**  $S^1 \times S^1 = \mathbb{R}^2/M$ , where  $M = \mathbb{Z}\lambda + \mathbb{Z}\tau$  is the period lattice obtained by integrating  $\int dx/y$  along loops. The torus carries a group law, the seed of 19th-century geometry and modern elliptic-curve cryptography.

? Beware two "tori": this *topological* torus is *not* the algebraic torus  $(\mathbb{C}^*)^n$  of toric geometry (Ch. 8). An elliptic curve is not toric.

### The dictionary, completed

Look back at the front-page table and check every row you can now fill. The two chapters were one conversation, in two languages:

| Algebra                                       | Geometry                                                        |
|-----------------------------------------------|-----------------------------------------------------------------|
| prime ideal                                   | irreducible variety (Prop. 4.3)                                 |
| maximal ideal                                 | a point                                                         |
| $I \subseteq J$                               | $\mathcal{V}(I) \supseteq \mathcal{V}(J)$ (inclusion-reversing) |
| $I + J$ / $I \cap J$                          | intersection / union                                            |
| dimension, degree of $I$                      | dimension, degree of $\mathcal{V}(I)$ (they agree!)             |
| homogeneous ideal                             | projective variety                                              |
| $\mathcal{I}(\mathcal{V}(I)) = \text{rad}(I)$ | the Nullstellensatz bridge (Ch. 6)                              |
| primary decomposition                         | decomposition into irreducibles (Ch. 3)                         |

Where this goes next. Chapter 3 (*Solving and Decomposing*) turns " $\mathcal{V}(I)$  = union of irreducibles" into an algorithm: **primary decomposition**, the geometry behind the Chinese Remainder Theorem you already met in the running cubic. Keep the cubic, the dictionary, and the Gröbner engine in hand; everything later is built from these three.

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Exercise 15 (book): solve Diophantus' problem -- find a rational point on the elliptic curve $y(6-y) = x^3 - x$. Hint: take the tangent line at $(-1, 0)$ and reintersect.